

POCKETSMART AI

Project Development Phase

Project Goal: PocketSmart AI is designed to help users manage their money in a simple and smart way. It can help with expense tracking, budgeting, savings guidance, and basic financial insights.

1. Requirement Analysis

- Identify the main problems users face while managing daily expenses.
- Collect requirements for income, expenses, budget, savings, and financial insights.
- Identify the target users and their common needs.
- Define the main features that should be included in the application.

2. System Design

The system is planned before coding. The user interface, application flow, database structure, and AI-related functions are designed in this phase.

- Design simple screens for adding and viewing expenses.
- Design the budget and savings sections.
- Plan how user data will be stored and retrieved.
- Plan how the AI component will process user information and provide useful insights.

3. Technology Selection

Component	Purpose
Python	Main programming language
Streamlit	Create the user interface
Gemini AI	AI-based analysis and assistance
Database / Storage	Store user expense and budget information
.env file	Keep API keys and sensitive configuration safe

4. Application Development

- Create the PocketSmart AI project folder and required files.
- Build the Streamlit interface.
- Create forms to enter income and expenses.
- Add budget and savings calculation features.
- Connect the application with the AI service.
- Display useful results in a clear and understandable format.

5. AI Integration

The AI part is used to understand the user's financial information and provide helpful suggestions. The application can generate simple insights such as spending patterns, budget suggestions, and savings guidance based on the information provided.

- Connect the application to the Gemini API.
- Send appropriate user information to the AI model.
- Receive the AI response.
- Display the response in a simple user-friendly way.

6. Testing Phase

- Test each feature separately.
- Check whether expense and budget calculations are correct.
- Test valid and invalid user inputs.
- Check whether the AI response is displayed correctly.
- Fix errors and improve the application.

7. Deployment and Finalization

- Run the complete application and verify all modules.
- Secure API keys and configuration details.
- Improve the user interface and remove unnecessary errors.
- Prepare the final project for demonstration.
- Document the project features, development process, and future improvements.

8. Development Flow

Requirement Analysis → System Design → Technology Selection → Application Development → AI Integration → Testing → Deployment

Conclusion

The development phase of PocketSmart AI converts the project idea into a working application. By combining a simple user interface, financial tracking features, and AI-based assistance, the system can provide users with understandable and useful support for managing their personal finances.